EDGE AI · FOOD SAFETY . YOLO .COMPUTER VISION

Real-Time Kitchen Hygiene and Safety Monitoring

Using YOLO and TensorRT on NVIDIA Jetson Orin Nano



TEAM

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Tech Stack

Trained on 7k+ images, then an FP16 TensorRT Quantaxtiom and running live inference on the Jetson Nano

TRAINING · NVIDIA DGX B200 (college gpu)

1. **Dataset**
   Mask, gloves, hairnet, cockroach plus “without” the violation classes
2. **Preprocessing**
   Labels cleaned & merged · resized to 640×640 · augmented
3. **Model training(YOLO11s)**
   9.4M params · 150 epochs · PyTorch 2.8 · Ultralytics 8.4
4. **Trained weights**
   best.pt · mAP@50 0.966 on 2,351 validation images

best.pt copied to the Jetson

EDGE DEPLOYMENT · JETSON ORIN NANO

5. **ONNX export**
   opset 12 · graph simplified p=9.34
6. **TensorRT FP16**
   trtexec · engine, built on the Jetson
7. **Live inference**
   USB webcam · OpenCV · CUDA · 15 FPS
8. **Violation rule**
   Any “without” class or cockroach = violation
9. **Dashboard**
   Class counts · FPS · timestamped alerts

Requirement.txt= JetPack 6.2.1 · TensorRT 10.3.0 · OpenCV 4.10 · NumPy 1.26.4

Key Stats & Highlights

Real-time kitchen hygiene monitoring on a low-cost edge board

0.966

mAP@50

Validation, 2,351 images

0.965

Precision

Few false alarms

0.957

Recall

Most violations caught

15 FPS

Live speed on nano

Camera + inference on Jetson

7.4 W

Avg. input power

No cloud server needed

17.9 MB

FP16 engine

TensorRT, built on-device

Project highlights

- **The problem:** hygiene is checked only now and then, so violations go unnoticed
- **What it detects:** mask, gloves, hairnet, their “without” versions and cockroaches
- **How:** YOLO11s trained on a merged public dataset, run as an FP16 TensorRT engine
- **Output:** live dashboard with violation count and timestamped alerts
- **Limits:** “no hairnet” is weakest (0.870 mAP@50); live cockroach confidence only 0.29

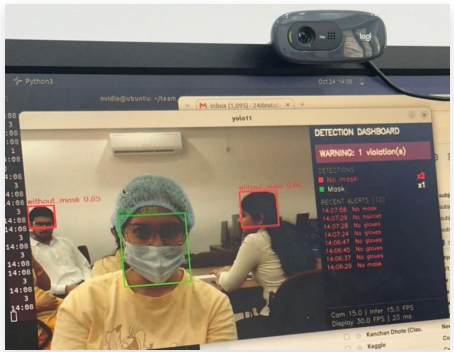
Live Outputs

Live webcam frames running on the Jetson Orin Nano



Compliance check

Mask, hairnet and glove detected; missing glove flagged



Mask vs no mask

No-mask violations flagged in red



Cockroach alert

Pest detected alongside a no-mask violation

Live violation count and timestamped alert log

Requirement met

Violation